动力学建模与仿真

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目录

第一章

广义变分原理

§1.1 多元函数极值问题的描述

1.1.1 多元函数

设 $\underline{x} = [x_1, x_2, \ldots, x_n]^T \in \mathbb{R}^n$，则关于变量 x_i $(i = 1, 2, \ldots, n)$ 的 n 元实函数记为

$$y = f(x_1, x_2, \ldots, x_n) \in \mathbb{R},$$

可简记为 $y = f(\underline{x})$。

1.1.2 多元函数的极值

【定义 1.1】(极小值与极小值点) 若多元函数 $f(\underline{x})$ 在 $\underline{x}_0$ 及其去心邻域内有定义，且对于所有满足

$$0 < \|\delta\underline{x}\| < \varepsilon$$

的自变量微小变更 $\delta\underline{x}$，都有

$$f(\underline{x}_0) \leq f(\underline{x}_0 + \delta\underline{x}),$$

则称 $\underline{x}_0$ 为函数 $f(\underline{x})$ 的一个 ** 极小值点 **，相应地，$f(\underline{x}_0)$ 为函数 $f(\underline{x})$ 的一个 ** 极小值 **。

【定义 1.2】(极大值与极大值点) 在相同条件下，如果都有

$$f(\underline{x}_0) \geq f(\underline{x}_0 + \delta\underline{x}),$$

则称 $\underline{x}_0$ 为函数 $f(\underline{x})$ 的一个 ** 极大值点 **，相应地，$f(\underline{x}_0)$ 为函数 $f(\underline{x})$ 的一个 ** 极大值 **。

<u>注.</u> 微小变更 $\delta\underline{x}$ 可不在乎方向。

1.1.3 多元函数的泰勒展开式

设多元函数 $y = f(\underline{x})$ 在点 $\underline{x}_0$ 的邻域内具有 $(m+1)$ 阶连续偏导数，则多元函数在该邻域内任一点 $(\underline{x}_0 + \delta\underline{x})$ 处的值可由 Taylor 展开式表示为（$0 < \theta < 1$）：

$$\begin{aligned} f(\underline{x}_0 + \delta\underline{x}) = {} & f(\underline{x}_0) + Df(\underline{x})\big|_{\underline{x}=\underline{x}_0} + \frac{1}{2!}D^2 f(\underline{x})\big|_{\underline{x}=\underline{x}_0} \\ & + \cdots + \frac{1}{m!}D^m f(\underline{x})\big|_{\underline{x}=\underline{x}_0} + \frac{1}{(m+1)!}D^{m+1} f(\underline{x})\big|_{\underline{x}=\underline{x}_0+\theta\delta\underline{x}}, \end{aligned}$$

其中算子 D 的表达式为

$$D = \left(\delta x_1 \frac{\partial}{\partial x_1} + \delta x_2 \frac{\partial}{\partial x_2} + \cdots + \delta x_n \frac{\partial}{\partial x_n}\right).$$

1.1.4 多元函数的驻点

多元函数 f 在点 $(\underline{x}_0 + \delta \underline{x})$ 处的一阶 Taylor 展开式为

$$\begin{aligned} f(\underline{x}_0 + \delta\underline{x}) &= f(\underline{x}_0) + Df(\underline{x})\big|_{\underline{x}=\underline{x}_0} + \frac{1}{2!}D^2 f(\underline{x})\big|_{\underline{x}=\underline{x}_0+\theta\delta\underline{x}} \\ &\approx f(\underline{x}_0) + Df(\underline{x})\big|_{\underline{x}=\underline{x}_0} \\ &= f(\underline{x}_0) + \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}_{\underline{x}=\underline{x}_0} \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \vdots \\ \delta x_n \end{bmatrix} \\ &=: f(\underline{x}_0) + f_{\underline{x}}(\underline{x}_0)\delta\underline{x}. \end{aligned}$$

若 $\underline{x}_0$ 为一个极小值点，则

$$f(\underline{x}_0 + \delta\underline{x}) - f(\underline{x}_0) + f_{\underline{x}}(\underline{x}_0)\delta\underline{x} \geq f(\underline{x}_0).$$

进而有

$$\forall \delta\underline{x}, \quad f_{\underline{x}}(\underline{x}_0)\delta\underline{x} \geq 0.$$

由于 $\delta\underline{x}$ 方向任意，故

$$f_{\underline{x}}(\underline{x}_0)\delta\underline{x} = 0 \implies f_{\underline{x}}(\underline{x}_0) = \underline{0}.$$

因此，$\underline{x}_0$ 为函数 f 的一个极小值点 $\implies$ $\underline{x}_0$ 为 f 的一个 ** 驻点 **。

§1.2 含冗余约束的非独立变分

设 $\underline{\alpha} \in \mathbb{R}^n$，$\underline{q} \in \mathbb{R}^n$，$\underline{\Phi} \in \mathbb{R}^{m\times n}$，且满足

$$\delta\underline{q}^T\underline{\alpha} = 0 \quad \forall\delta\underline{q} : \ \underline{\Phi}\delta\underline{q} = \underline{0}.$$

则存在一个 Lagrange 乘子 $\underline{\lambda} \in \mathbb{R}^m$，满足

$$\underline{\alpha} + \underline{\Phi}^T\underline{\lambda} = \underline{0}.$$

即

$$\delta\underline{q}^T\underline{\alpha} = 0 \quad \forall\delta\underline{q} : \ \underline{\Phi}\delta\underline{q} = \underline{0} \implies \exists\underline{\lambda} \text{ s.t. } \underline{\alpha} + \underline{\Phi}^T\underline{\lambda} = \underline{0}.$$

证明. 通过行初等变换矩阵 R、C，可将 $\underline{\Phi}$ 变换为

$$R\underline{\Phi}C = \begin{bmatrix} \underline{\Phi}_D & \underline{\Phi}_I \\ \underline{O} & \underline{O} \end{bmatrix} =: \begin{bmatrix} \overline{\underline{\Phi}} \\ \underline{O} \end{bmatrix}.$$

则

$$\underline{\Phi}\delta\underline{q} = \underline{0} \iff R\underline{\Phi}CC^{-1}\delta\underline{q} = \underline{0} \iff \overline{\underline{\Phi}}C^{-1}\delta\underline{q} = \underline{0}.$$

定义 $\delta\overline{\underline{q}} := C^{-1}\delta\underline{q}$，即 $\delta\underline{q} = C\delta\overline{\underline{q}}$。则

$$0 = \delta\underline{q}^T\underline{\alpha} = (C\delta\overline{\underline{q}})^T\underline{\alpha} = \delta\overline{\underline{q}}^T C^T\underline{\alpha}.$$

即

$$0 = \delta\overline{\underline{q}}^T C^T\underline{\alpha} \quad \forall\delta\overline{\underline{q}} :\ \overline{\underline{\Phi}}\delta\overline{\underline{q}} = \underline{0}.$$

将 $\overline{\underline{\Phi}} = [\underline{\Phi}_D\ \underline{\Phi}_I] \in \mathbb{R}^{r\times n}$（$r \le n$，$\underline{\Phi}_D \in \mathbb{R}^{r\times r}$）分块，则存在 $\underline{\lambda} \in \mathbb{R}^r$，使得

$$C^T\underline{\alpha} + \overline{\underline{\Phi}}^T\underline{\lambda} = \underline{0}.$$

引入 $\underline{\lambda}_0 \in \mathbb{R}^{m-r}$，则

$$C^T\underline{\alpha} + \overline{\underline{\Phi}}^T\underline{\lambda} + \underline{O}^T\underline{\lambda}_0 = \underline{0}.$$

令 $\underline{\Lambda} = R^T\begin{bmatrix} \underline{\lambda} \\ \underline{\lambda}_0 \end{bmatrix}$，则

$$\begin{aligned} \underline{0} &= C^T\underline{\alpha} + \begin{bmatrix} \overline{\underline{\Phi}} \\ \underline{O} \end{bmatrix}^T \begin{bmatrix} \underline{\lambda} \\ \underline{\lambda}_0 \end{bmatrix} \\ &= C^T\underline{\alpha} + (R\underline{\Phi}C)^T \begin{bmatrix} \underline{\lambda} \\ \underline{\lambda}_0 \end{bmatrix} \\ &= C^T\underline{\alpha} + C^T\underline{\Phi}^T R^T \begin{bmatrix} \underline{\lambda} \\ \underline{\lambda}_0 \end{bmatrix} \\ \implies \underline{0} &= \underline{\alpha} + \underline{\Phi}^T R^T \begin{bmatrix} \underline{\lambda} \\ \underline{\lambda}_0 \end{bmatrix} = \underline{\alpha} + \underline{\Phi}^T\underline{\Lambda}. \end{aligned}$$

□

§1.3 多元函数条件极值问题求解

约束方程：

$$\underline{0} = C_{\underline{x}}(\underline{x})\delta\underline{x}.$$

极值条件：

$$f_{\underline{x}}(\underline{x})\delta\underline{x} = 0.$$

由广义变分原理，得

$$\begin{cases} f_{\underline{x}}^T(\underline{x}_0) + C_{\underline{x}}^T(\underline{x}_0)\underline{\lambda} = \underline{0}, \\ f_{\underline{x}}(\underline{x}_0) + C_{\underline{x}}^T(\underline{x}_0)\underline{\lambda} = \underline{0}, \\ C(\underline{x}_0) = \underline{0}. \end{cases}$$

由此解出极值点。

第二章

拉格朗日方程

§2.1 系统广义坐标和约束方程

n 维广义坐标用 n 维列向量表示：

$$\underline{q}(t) = [q_1(t)\ q_2(t)\ \ldots\ q_n(t)]^T.$$

这里要求广义坐标独立，即 $n = \text{DOFs}$。则编号为 k 的粒子，其全局坐标可表示为

$$\vec{r}_k(q_1(t), q_2(t), \ldots, q_n(t)) =: \vec{r}_k(\underline{q}(t)).$$

由此我们有系统广义速度：

$$\begin{aligned}\frac{d}{dt}\vec{r}_k(\underline{q}(t)) &= \sum_{i=1}^{n} \frac{\partial \vec{r}_k}{\partial q_i}\frac{dq_i}{dt} \\ &= \sum_{i=1}^{n} \frac{\partial \vec{r}_k}{\partial q_i}\dot{q}_i \\ &=: \dot{\vec{r}}_k(\underline{q}(t), \underline{\dot{q}}(t)).\end{aligned}$$

若 $\dim(\vec{r}_k) > \text{DOFs}$，则存在约束方程

$$0 = C_i(\underline{q}(t), t) \quad (i = 1, 2, \ldots, m).$$

可写作向量形式

$$\vec{0} = \vec{C}(\underline{q}(t), t).$$

下面我们来讨论约束方程对时间的导数：

$$\begin{aligned}0 &= C_i(\underline{q}(t), t), \\ 0 &= \sum_{j=1}^{n} \frac{\partial C_i}{\partial q_j}\frac{\partial q_j}{\partial t} + \frac{\partial C_i}{\partial t} \\ &= \sum_{j=1}^{n} \frac{\partial C_i}{\partial q_j}\dot{q}_j + \frac{\partial C_i}{\partial t} \\ &=: \dot{C}_i(\underline{q}(t), \underline{\dot{q}}(t), t) \quad (i = 1, 2, \ldots, m).\end{aligned}$$

若令 $\vec{C}_{\underline{q}}(\underline{q}(t),t)$（约束方程的雅各比矩阵）：

$$\vec{C}_{\underline{q}}(\underline{q}(t),t)=\begin{bmatrix}\frac{\partial C_1}{\partial q_1} & \frac{\partial C_1}{\partial q_2} & \cdots & \frac{\partial C_1}{\partial q_n}\\ \frac{\partial C_2}{\partial q_1} & \frac{\partial C_2}{\partial q_2} & \cdots & \frac{\partial C_2}{\partial q_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial C_m}{\partial q_1} & \frac{\partial C_m}{\partial q_2} & \cdots & \frac{\partial C_m}{\partial q_n}\end{bmatrix}\in\mathbb{R}^{m\times n},$$

$$\vec{C}_t(\underline{q}(t),t)=\begin{bmatrix}\frac{\partial C_1}{\partial t}\\ \frac{\partial C_2}{\partial t}\\ \vdots\\ \frac{\partial C_m}{\partial t}\end{bmatrix}\in\mathbb{R}^{m\times 1},$$

则有广义速度层面的约束方程：

$$\vec{0}=\vec{C}_{\underline{q}}(\underline{q}(t),t)\dot{\underline{q}}(t)+\vec{C}_t(\underline{q}(t),t).$$

2.1.1 广义坐标的微小变更

广义坐标的约束方程：

$$0=C_i(\underline{q}(t),t)=:C_i(\underline{q}(t),t).$$

引入一个广义虚位移 $\delta\underline{q}(t)$，

$$0=C_i(\underline{q}+\delta\underline{q},t).$$

进行一阶 Taylor 展开：

$$\begin{aligned}0=C_i(\underline{q}+\delta\underline{q},t)&\approx C_i(\underline{q},t)+\sum_{j=1}^{n}\frac{\partial C_i(\underline{q},t)}{\partial q_j}\delta q_j\\ &=\sum_{j=1}^{n}\frac{\partial C_i(\underline{q},t)}{\partial q_j}\delta q_j\quad(j=1,2,\dots,n).\end{aligned}$$

则

$$\vec{0}=\begin{bmatrix}\frac{\partial C_1}{\partial q_1} & \frac{\partial C_1}{\partial q_2} & \cdots & \frac{\partial C_1}{\partial q_n}\\ \frac{\partial C_2}{\partial q_1} & \frac{\partial C_2}{\partial q_2} & \cdots & \frac{\partial C_2}{\partial q_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial C_m}{\partial q_1} & \frac{\partial C_m}{\partial q_2} & \cdots & \frac{\partial C_m}{\partial q_n}\end{bmatrix}\begin{bmatrix}\delta q_1\\ \delta q_2\\ \vdots\\ \delta q_n\end{bmatrix}=:\vec{C}_{\underline{q}}(\underline{q},t)\delta\underline{q}=\frac{\partial\vec{C}(\underline{q},t)}{\partial\underline{q}}\delta\underline{q}.$$

2.1.2 质点的虚位移

$$\vec{r}_k=\vec{r}_k(\underline{q}(t)),\quad \vec{r}_k(\underline{q}+\delta\underline{q}),$$

$$\delta\vec{r}_k=\vec{r}_k(\underline{q}+\delta\underline{q})-\vec{r}_k(\underline{q}).$$

由一阶 Taylor 展开：

$$\delta\vec{r}_k \approx \sum_{j=1}^{n} \frac{\partial \vec{r}_k}{\partial q_j}\delta q_j,$$

即

$$\delta\vec{r}_k = \sum_{j=1}^{n} \frac{\partial \vec{r}_k}{\partial q_j}\delta q_j.$$

§2.2 虚功原理与拉格朗日方程

2.2.1 质点的运动微分方程

Newton 第二运动定律：

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = \frac{d}{dt}\left(m\frac{d\vec{r}}{dt}\right) = \frac{dm}{dt}\frac{d\vec{r}}{dt} + m\frac{d^2\vec{r}}{dt^2}.$$

一个具有恒定质量的物体（不考虑相对论效应），则 $\frac{dm}{dt} = 0$，

$$\vec{F} = m\frac{d^2\vec{r}}{dt^2} =: m\ddot{\vec{r}} =: m\vec{a}.$$

若将 $(-m\ddot{\vec{r}})$ 视作惯性力（Inertial force），则有 D'Alembert 原理（达朗贝尔原理）：

$$-m\ddot{\vec{r}} + \vec{F} = \vec{0}.$$

对于一个由 k 个部分组成的多体系统：

$$-m_k\ddot{\vec{r}}_k + \vec{F}_{\mathrm{EX},k} + \vec{F}_{\mathrm{IN},k} = \vec{0}.$$

2.2.2 质点与质点系的虚功方程

对于质点系中的第 k 个质点，有动力学方程

$$-m_k\ddot{\vec{r}}_k + \vec{F}_{\mathrm{EX},k} + \vec{F}_{\mathrm{IN},k} = \vec{0}.$$

由此有虚功方程

$$-m_k\delta\vec{r}_k^T\ddot{\vec{r}}_k + \delta\vec{r}_k^T\vec{F}_{\mathrm{EX},k} + \delta\vec{r}_k^T\vec{F}_{\mathrm{IN},k} = \vec{0}.$$

将质点系中所有的方程相加：

$$-\sum_{k=1}^{N} m_k\delta\vec{r}_k^T\ddot{\vec{r}}_k + \sum_{k=1}^{N}\delta\vec{r}_k^T\vec{F}_{\mathrm{EX},k} + \sum_{k=1}^{N}\delta\vec{r}_k^T\vec{F}_{\mathrm{IN},k} = \vec{0}.$$

可以看到，虚功方程由三部分组成：

1. 惯性力的虚功：$-\sum_{k=1}^{N} m_k \delta \vec{r}_k^T \ddot{\vec{r}}_k$

2. 系统外力的虚功：$\sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{EX},k}$

3. 系统内力的虚功：$\sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{IN},k}$

下面逐一剖析：

系统惯性力的虚功

$$
\begin{aligned}
-\sum_{k=1}^{N} m_k \delta \vec{r}_k^T \ddot{\vec{r}}_k &= -\sum_{k=1}^{N} m_k \left(\sum_{j=1}^{n} \frac{\partial \vec{r}_k}{\partial q_j} \delta q_j \right)^T \frac{d}{dt} \left(\dot{\vec{r}}_k \right) \\
&= -\sum_{k=1}^{N} m_k \sum_{j=1}^{n} \delta q_j \frac{\partial \vec{r}_k^T}{\partial q_j} \frac{d}{dt} \left(\dot{\vec{r}}_k \right) \\
&= -\sum_{j=1}^{n} \sum_{k=1}^{N} m_k \delta q_j \frac{\partial \vec{r}_k^T}{\partial q_j} \frac{d}{dt} \left(\dot{\vec{r}}_k \right) \\
&= -\sum_{j=1}^{n} \delta q_j \sum_{k=1}^{N} m_k \left[\frac{d}{dt} \left(\frac{\partial \vec{r}_k^T}{\partial q_j} \dot{\vec{r}}_k \right) - \frac{d}{dt} \left(\frac{\partial \vec{r}_k^T}{\partial q_j} \right) \dot{\vec{r}}_k \right].
\end{aligned}
$$

注意到

$$
\dot{\vec{r}}_k = \frac{d\vec{r}_k}{dt} = \sum_{j=1}^{n} \frac{\partial \vec{r}_k}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_k}{\partial t},
$$

$$
\frac{\partial \dot{\vec{r}}_k}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j} \left(\sum_{j=1}^{n} \frac{\partial \vec{r}_k}{\partial q_j} \dot{q}_j \right) = \frac{\partial \vec{r}_k}{\partial q_j},
$$

即

$$
\frac{\partial \dot{\vec{r}}_k}{\partial \dot{q}_j} = \frac{\partial \vec{r}_k}{\partial q_j}.
$$

又

$$
\frac{d}{dt} \left(\frac{\partial \vec{r}_k}{\partial q_j} \right) = \frac{\partial \dot{\vec{r}}_k}{\partial q_j}.
$$

于是

$$
\begin{aligned}
\sum_{k=1}^{N} m_k \frac{\partial \vec{r}_k^T}{\partial q_j} \dot{\vec{r}}_k &= \sum_{k=1}^{N} m_k \frac{\partial \dot{\vec{r}}_k^T}{\partial \dot{q}_j} \dot{\vec{r}}_k \\
&= \sum_{k=1}^{N} m_k \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} \dot{\vec{r}}_k^T \dot{\vec{r}}_k \right) \\
&= \frac{\partial}{\partial \dot{q}_j} \left(\sum_{k=1}^{N} \frac{1}{2} m_k \dot{\vec{r}}_k^T \dot{\vec{r}}_k \right) = \frac{\partial T}{\partial \dot{q}_j}.
\end{aligned}
$$

因此

$$
\begin{aligned}
-\sum_{k=1}^{N} m_k \delta \vec{r}_k^T \ddot{\vec{r}}_k &= -\sum_{j=1}^{n} \delta q_j \left[\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} \right] \\
&= \sum_{j=1}^{n} \delta q_j \left[-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) + \frac{\partial T}{\partial q_j} \right].
\end{aligned}
$$

系统内力与外力的虚功

$$
\begin{aligned}
\sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{EX},k} + \sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{IN},k} &= \sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{EX},k} + \sum_{k=1}^{N} \delta \vec{r}_k^T \vec{F}_{\mathrm{NonIdeal},k} \\
&= \sum_{k=1}^{N} \delta \vec{r}_k^T \left(\vec{F}_{\mathrm{EX},k} + \vec{F}_{\mathrm{NonIdeal},k} \right) \\
&= \sum_{k=1}^{N} \sum_{j=1}^{n} \delta q_j \frac{\partial \vec{r}_k^T}{\partial q_j} \left(\vec{F}_{\mathrm{EX},k} + \vec{F}_{\mathrm{NonIdeal},k} \right) \\
&= \sum_{j=1}^{n} \delta q_j \sum_{k=1}^{N} \frac{\partial \vec{r}_k^T}{\partial q_j} \left(\vec{F}_{\mathrm{EX},k} + \vec{F}_{\mathrm{NonIdeal},k} \right) \\
&=: \sum_{j=1}^{n} \delta q_j Q_j.
\end{aligned}
$$

于是，虚功方程被简化为：

$$
0 = \sum_{j=1}^{n} \delta q_j \left[-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) + \frac{\partial T}{\partial q_j} + Q_j \right].
$$

写成矩阵形式：

$$
0 = \delta \underline{q}^T \left[-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\underline{q}}}\right)^T + \left(\frac{\partial T}{\partial \underline{q}}\right)^T + \vec{Q} \right],
$$

其中 $\delta \underline{q}$ 满足 $\underline{0} = \frac{\partial \vec{C}}{\partial \underline{q}} \delta \underline{q}$。由广义变分原理，$\exists \underline{\lambda} \in \mathbb{R}^m$，s.t.

$$
-\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\underline{q}}}\right)^T + \left(\frac{\partial T}{\partial \underline{q}}\right)^T + \vec{Q} - \left(\frac{\partial \vec{C}}{\partial \underline{q}}\right)^T \underline{\lambda} = \underline{0}.
$$

§2.3 系统动能与广义质量矩阵

2.3.1 系统动能 (The system kinetic energy)

【定义 2.1】(系统动能)

$$T=\sum_{k=1}^{N}\frac{1}{2}m_k\dot{\vec{r}}_k^T\dot{\vec{r}}_k.$$

这里，

$$\dot{\vec{r}}_k=\sum_{j=1}^{n}\frac{\partial\vec{r}_k}{\partial q_j}\frac{dq_j}{dt}=\sum_{j=1}^{n}\frac{\partial\vec{r}_k}{\partial q_j}\dot{q}_j=:\vec{r}_{k,\underline{q}}\dot{\underline{q}}.$$

则

$$\begin{aligned}T&=\sum_{k=1}^{N}\frac{1}{2}m_k\left(\vec{r}_{k,\underline{q}}\dot{\underline{q}}\right)^T\left(\vec{r}_{k,\underline{q}}\dot{\underline{q}}\right)\\&=\sum_{k=1}^{N}\frac{1}{2}m_k\dot{\underline{q}}^T\vec{r}_{k,\underline{q}}^T\vec{r}_{k,\underline{q}}\dot{\underline{q}}\\&=\frac{1}{2}\dot{\underline{q}}^T\left(\sum_{k=1}^{N}m_k\vec{r}_{k,\underline{q}}^T\vec{r}_{k,\underline{q}}\right)\dot{\underline{q}}.\end{aligned}$$

2.3.2 系统广义质量矩阵 (The generalized inertial matrix)

【定义 2.2】(广义质量矩阵)

$$\vec{M}(\underline{q})=\sum_{k=1}^{N}m_k\vec{r}_{k,\underline{q}}^T\vec{r}_{k,\underline{q}}.$$

则系统动能

$$T=\frac{1}{2}\dot{\underline{q}}^T\vec{M}(\underline{q})\dot{\underline{q}}.$$

【命题 2.3】(广义质量矩阵的性质) 1. 对称性 **(Symmetry)**：$\vec{M}=\vec{M}^T$。

证明.

$$M_{jk}=\sum_{i=1}^{N}m_i\frac{\partial\vec{r}_i}{\partial q_j}\cdot\frac{\partial\vec{r}_i}{\partial q_k},\quad M_{kj}=\sum_{i=1}^{N}m_i\frac{\partial\vec{r}_i}{\partial q_k}\cdot\frac{\partial\vec{r}_i}{\partial q_j}.$$

显然 $M_{jk}=M_{kj}$，故 $\vec{M}=\vec{M}^T$。 □

2. 正定性 **(Positive Definiteness)**：$\vec{M}$ 始终正定或半正定。

证明. $T\geq 0\implies 2T\geq 0$，即

$$\dot{\underline{q}}^T\vec{M}(\underline{q})\dot{\underline{q}}\geq 0.$$

故 $\vec{M}(\underline{q})$ 正定或半正定。 □

注. 存在奇异位姿，即 $|\vec{M}|=0$ 时，$\vec{M}(\underline{q})$ 半正定。